

Resolving Stable Boundary Layer Transitions via Geometric Singular Perturbation Theory: Isolating Coordinate Fold Illusions from Turbulence Extinction

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Abstract

The transition between Weakly Stable (WSBL) and Very Stable Boundary Layer (VSBL) regimes is frequently obfuscated by visual gradient inflections (“knees”) and numerical instabilities near critical gradient Richardson thresholds ($Ri_c \approx 0.25$) [? ?]. This paper presents an analytical framework grounded in Geometric Singular Perturbation Theory (GSPT) that mathematically decouples physical turbulence collapse from non-linear, observation-space coordinate projection geometry [? ?]. By embedding a velocity-scale regularization floor ($\delta = 10^{-6} \text{ m}^2\text{s}^{-2}$) into the microscale TKE equation, C^1 -continuity of the fast vector field is restored across zero-energy limits, stabilizing linear stability analysis near the laminar state [? ?]. We decompose total physical vertical spatial curvature of the Richardson profile into constitutive and mapping components ($Ri_{zz} = C_{\text{constitutive}} + C_{\text{mapping}}$) [? ?] and prove via the *Linear Obukhov Profile Test* that finite coordinate folds ($\zeta_z = 0$) cannot exist under linear Obukhov structures, requiring instead non-zero vertical curvature in the Obukhov length profile ($L''(z) \neq 0$) [? ? ?]. Near Low-Level Jet (LLJ) cores where velocity shear vanishes ($U_z \rightarrow 0$), mapping curvature asymptotically dominates ($C_{\text{mapping}} \gg C_{\text{constitutive}}$), manufacturing visual “Fold Illusions” distinct from true state-space saddle-node bifurcations ($\det D_X F = 0$ subject to standard nondegeneracy and transversality criteria) [? ? ?]. This GSPT structure is operationalized on 200 m Cabauw mast soundings (GABLS3) using a campaign-portable NetCDF ingestion engine, where we regularize primitive fields prior to differentiation to bypass non-commutation operator artifacts [? ? ?]. Finally, we establish a closed spatial curvature residual partition ($\Delta_{\text{SCM}} = \Delta_{\text{closure}} + \Delta_{\text{geom}} + \tau_{\Delta z}$) that isolates constitutive parameterization failures from grid-resolution and flux-coordinate distortions [? ? ?]. Ultimately, this framework demonstrates that the classical threshold crossing $Ri_g = Ri_{\text{critical}}$ is neither a coordinate-fold criterion nor a dynamical saddle-node criterion [?].

1 Introduction

The representation of the stable atmospheric boundary layer (SBL) in numerical weather prediction (NWP) and single-column models (SCMs) remains one of the most enduring challenges in boundary-layer meteorology [?]. Under stable stratification, the vertical structure of the boundary layer is governed by a delicate, highly non-linear balance among mechanical shear production, buoyant consumption, and molecular dissipation. SBLs are conventionally categorized into two distinct physical regimes: the Weakly Stable Boundary Layer (WSBL), characterized by continuous mechanical shear production and robust vertical coupling with the surface, and the Very Stable Boundary Layer (VSBL), where turbulent mixing is strongly suppressed, leading

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to surface-radiation decoupling, rapid land-surface temperature drops, and strong nocturnal ground inversions [? ?].

Historically, the transition between these two regimes has been diagnosed using local stability parameters, most notably the gradient Richardson number (Ri_g). The classical transition is assumed to occur at a universal critical threshold ($Ri_c \approx 0.25$), above which mechanical shear can no longer sustain vertical turbulent fluxes, resulting in an abrupt “turbulent collapse” or Monin–Obukhov Similarity Theory (MOST) breakdown [?]. However, field campaigns (such as CASES-99 and SHEBA) and high-resolution Large-Eddy Simulations (LES) demonstrate that SBL transitions are highly non-stationary, exhibiting significant hysteresis, intermittency, and a distinct lack of a universal critical Richardson number [? ?]. This lack of universality arises because a single scalar Richardson number represents a projection of a higher-dimensional dynamical state; different atmospheric columns reach the same physical transition at vastly different apparent values of Ri_g .

In this paper, we establish an analytical and geometric framework grounded in Geometric Singular Perturbation Theory (GSPT) to analyze SBL transitions. We model the stable boundary layer as a coupled slow-fast dynamical system, wherein the microscale Turbulent Kinetic Energy (TKE), E , adjusts rapidly on a fast timescale, while the mesoscale vertical wind shear, S , evolves slowly under large-scale geostrophic forcing. To resolve the numerical singularities that plague standard unregularized boundary layer formulations near the laminar state ($E \rightarrow 0$), we introduce a velocity-scale regularization floor ($\delta = 10^{-6} \text{ m}^2\text{s}^{-2}$) into the governing equations, restoring C^1 -continuity of the fast vector field across zero-energy limits.

Furthermore, we outline the Fenichel slow-manifold invariance equations to characterize the quasi-equilibrium states where turbulence “rides” a slow manifold, proving that regime transitions correspond to a loss of normal hyperbolicity at nondegenerate state-space saddle-node folds ($\det D_X F = 0$) rather than unphysical scalar threshold crossings. Crucially, this state-space bifurcation is decoupled from observation-space kinematic folds ($\zeta_z = 0$), resolving visual gradient artifacts near Low-Level Jet (LLJ) cores.

1.1 Fast-Slow SBL System Formulation

The multiscale nature of SBL turbulence is mathematically formalized by parameterizing the timescale separation between fast microscale turbulence and slow mesoscale wind shear using a small dimensionless ratio $0 < \epsilon \ll 1$. Let $E \geq 0$ denote the turbulent kinetic energy per unit mass (m^2s^{-2}), and let $S \geq 0$ denote the vertical wind shear magnitude (s^{-1}). The prognostic equations for the coupled zero-dimensional SBL TKE budget are given by:

$$\epsilon \frac{dE}{dt} = f(E, S; N^2) = \mathcal{P}(E, S) - \mathcal{B}_{\text{bg}}(E, N^2) - \mathcal{B}_{\text{mod}}(E, N^2) - \varepsilon(E), \quad (1)$$

$$\frac{dS}{dt} = g(E, S) = G - c_1 ES - c_2 S, \quad (2)$$

where $N^2 > 0$ is the squared Brunt–Väisälä frequency (s^{-2}), representing ambient atmospheric stability, and $G > 0$ is the geostrophic forcing driving the wind speed aloft (s^{-2}).

Each physical component in the fast TKE budget (1) is closed using 1.5-order turbulence schemes where vertical transport is parameterized via an eddy viscosity $K_m = l\sqrt{E + \delta}$:

- **Mechanical Shear Production ($\mathcal{P}$):** Represents the extraction of energy from the mean velocity gradient by turbulent eddies, scaled by the mixing length l (m):

$$\mathcal{P}(E, S) = l\sqrt{E + \delta} S^2, \quad (3)$$

where a small regularization scale $\delta = 10^{-6} \text{ m}^2\text{s}^{-2}$ is added to the velocity-scale denominator of the production Jacobian term. In classical unregularized formulations ($\delta = 0$), the TKE

production term contains $\sqrt{E}$, yielding a singular Jacobian derivative at zero energy:

$$\lim_{E \rightarrow 0^+} \frac{\partial}{\partial E} (l\sqrt{E}S^2) = +\infty. \quad (4)$$

This singularity triggers severe numerical instabilities and prevents standard linear stability analysis near the laminar state. The regularization floor δ ensures that the fast vector field $f(E, S; N^2)$ remains C^1 -continuous on the physically admissible domain $\mathcal{D} = \{(E, S) \in \mathbb{R}^2 : E \geq 0\}$ while preserving dimensional consistency (m^2s^{-3}).

- **Background Buoyant Loss ($\mathcal{B}_{\text{bg}}$):** Parameterizes the conventional stable-stratification penalty associated with the positive mean potential-temperature gradient, governed by a background stability coefficient ϕ :

$$\mathcal{B}_{\text{bg}}(E, N^2) = l\sqrt{E + \delta} \phi N^2. \quad (5)$$

Because $\mathcal{B}_{\text{bg}}$ is strictly monotonic in E , it cannot by itself generate a manifold fold or regime transition.

- **Non-Monotonic Buoyant Destruction ($\mathcal{B}_{\text{mod}}$):** To capture wave-turbulence interactions and turbulent flux suppression observed in the VSBL, we introduce a non-monotonic heat flux saturation function $D_b(E; N^2)$ scaled directly by N^2 and a buoyant sink efficiency factor c_b :

$$\mathcal{B}_{\text{mod}}(E, N^2) = l\sqrt{E + \delta} D_b(E; N^2) = l\sqrt{E + \delta} c_b N^2 \frac{\alpha E}{(E + \alpha)^2}, \quad (6)$$

where $\alpha > 0$ is the regularized energy scale (m^2s^{-2}) at which buoyant destruction efficiency peaks. The derivative of this modified closure is:

$$D'_b(E) = c_b N^2 \alpha \frac{\alpha - E}{(E + \alpha)^3}, \quad (7)$$

which is positive for $E < \alpha$, zero at $E = \alpha$, and negative for $E > \alpha$. This change in sign of $D'_b(E)$ provides the necessary non-monotonicity in the fast vector field to permit the SBL equilibrium manifold to fold, a behavior that standard monotonic similarity closures (e.g., Businger–Dyer or Louis functions) fail to capture.

- **Viscous Dissipation (ε):** Parameterized via the conventional 1.5-order Kolmogorov-type dissipation rate:

$$\varepsilon(E) = \frac{E^{3/2}}{l}. \quad (8)$$

In the slow equation (2), geostrophic shear forcing G is balanced by background relaxation (with coefficient $c_2 \in \text{s}^{-1}$) and turbulent momentum mixing (with damping coefficient $c_1 \in \text{s}\cdot\text{m}^{-2}$). This forms a closed feedback loop:

$$S \longrightarrow \text{TKE Production } (\mathcal{P}) \longrightarrow E \longrightarrow \text{Turbulent Momentum Mixing } (c_1 ES) \longrightarrow S.$$

1.2 Fenichel Slow-Manifold Invariance Equations

In GSPT, the singular limit $\epsilon \rightarrow 0$ collapses the fast TKE dynamics onto the critical manifold $\mathcal{S}_0$, defined as the set of quasi-steady turbulent equilibria for a given shear state:

$$\mathcal{S}_0 = \{(E, S) \in \mathcal{D} : f(E, S; N^2) = 0\}. \quad (9)$$

On $\mathcal{S}_0$, the local relaxation rate of trajectories in the fast direction is governed by the fast eigenvalue $\lambda_f(E, S) \equiv \frac{\partial f}{\partial E}$:

$$\lambda_f(E, S) = \frac{l(S^2 - \phi N^2)}{2\sqrt{E + \delta}} - c_b N^2 l \left[\frac{E}{2\sqrt{E + \delta}(E + \alpha)} + \frac{\alpha\sqrt{E + \delta}}{(E + \alpha)^2} \right] - \frac{3\sqrt{E}}{2l}. \quad (10)$$

Under Fenichel's Theorem, any compact, normally hyperbolic branch of the critical manifold—where the fast eigenvalue remains bounded away from zero ($\lambda_f \neq 0$)—persists for sufficiently small $\epsilon > 0$ as a smooth, invariant slow manifold $\mathcal{S}_\epsilon$ [?]. This slow manifold is $\mathcal{O}(\epsilon)$ -close to $\mathcal{S}_0$ and completely governs the long-term, coupled evolution of the boundary layer state.

To mathematically define the geometry of the invariant slow manifold, we parameterize $\mathcal{S}_\epsilon$ as a function of the slow variable: $E = \Phi(S; \epsilon)$. Substituting this parameterization into the fast prognostic relation (1) and applying the chain rule yields the *Fenichel slow-manifold invariance equation*:

$$\epsilon \frac{\partial \Phi(S; \epsilon)}{\partial S} \frac{dS}{dt} = f(\Phi(S; \epsilon), S; N^2). \quad (11)$$

Substituting the slow shear tendency (2) into (11) yields the closed algebraic-differential invariance relation:

$$\epsilon \frac{\partial \Phi(S; \epsilon)}{\partial S} [G - c_1 \Phi(S; \epsilon) S - c_2 S] = f(\Phi(S; \epsilon), S; N^2). \quad (12)$$

Because $\mathcal{S}_\epsilon$ is $\mathcal{O}(\epsilon)$ -close to $\mathcal{S}_0$, we express the slow manifold parameterization as a regular perturbation expansion in powers of ϵ :

$$\Phi(S; \epsilon) = \Phi_0(S) + \epsilon \Phi_1(S) + \epsilon^2 \Phi_2(S) + \mathcal{O}(\epsilon^3). \quad (13)$$

Substituting (13) into the closed invariance relation (12) and equating powers of ϵ systematically resolves the manifold's geometry:

- **At $\mathcal{O}(1)$ (singular limit):**

$$f(\Phi_0(S), S; N^2) = 0. \quad (14)$$

This confirms that the leading-order term $\Phi_0(S)$ is exactly the critical manifold $\mathcal{S}_0$, representing instantaneous turbulent equilibrium.

- **At $\mathcal{O}(\epsilon)$ (first-order correction):**

$$\frac{\partial \Phi_0(S)}{\partial S} [G - c_1 \Phi_0(S) S - c_2 S] = \frac{\partial f(\Phi_0(S), S; N^2)}{\partial E} \Phi_1(S). \quad (15)$$

Solving for the first-order slow-manifold perturbation $\Phi_1(S)$ yields:

$$\Phi_1(S) = [\lambda_f(\Phi_0(S), S)]^{-1} \frac{\partial \Phi_0(S)}{\partial S} g(\Phi_0(S), S), \quad (16)$$

where λ_f is the local fast eigenvalue defined in (10).

Equation (16) highlights the crucial role of the fast eigenvalue in slow-manifold tracking. In normally hyperbolic regions where the attracting branch is stable ($\lambda_f < 0$), the first-order correction $\Phi_1(S)$ is bounded, and the physical trajectory rapidly relaxes onto $\mathcal{S}_\epsilon$ on a fast timescale $\mathcal{O}(\epsilon/|\lambda_f|)$ [?].

However, Fenichel's normal hyperbolicity condition breaks down entirely at candidate fold boundaries where the fast eigenvalue vanishes ($\lambda_f = f_E = 0$). At these points, the inverse eigenvalue in (16) diverges, and the slow manifold $\mathcal{S}_\epsilon$ ceases to exist. This breakdown marks the transition threshold where a slowly cooling boundary layer can no longer track the stable WSBL branch. Instead, the trajectory encounters a dynamic saddle-node fold and undergoes a fast-timescale jump across the unstable repelling branch ($\lambda_f > 0$) toward the low-energy attracting VSBL branch, providing a state-space explanation for abrupt SBL regime transitions and hysteresis.

2 Mathematical Curvature Decomposition

A central tenet of boundary-layer meteorology is that vertical profile shapes directly reflect the underlying physics of turbulent transport. However, when vertical profiles of gradient Richardson numbers (Ri_g) are plotted in physical vertical space (z), visual features such as sharp inflection points (“knees”) or threshold-crossing layers are frequently assumed to represent physical state-space bifurcations [?]. In this section, we present the mathematical foundation of Generalized Similarity Profile Theory (GSPT) to demonstrate that vertical profile morphology is a coupled product of the underlying turbulence closure and the differential geometry of the coordinate transformation [?]. By decoupling these effects, we show that visual knees can arise as pure coordinate-mapping features (referred to here as “Fold Illusions”) rather than dynamical bifurcations of the Turbulent Kinetic Energy (TKE) manifold [?].

2.1 The GSPT Curvature Decomposition

Let z denote physical vertical height (m), and let $L(z)$ represent the height-varying Monin–Obukhov length (m), which dynamically parameterizes local thermodynamic stability as a function of height-varying turbulent fluxes [?]. The non-dimensional Monin–Obukhov similarity coordinate $\zeta(z)$ is defined by the mapping:

$$\zeta(z) = \frac{z}{L(z)}. \quad (17)$$

To evaluate how vertical gradients and curvatures are projected from similarity space into physical observation space, we compute the first two spatial derivatives of the coordinate mapping $\zeta(z)$ with respect to physical height z . Applying the quotient rule directly to (17) yields the coordinate Jacobian $\zeta_z \equiv \frac{\partial \zeta}{\partial z}$:

$$\zeta_z = \frac{L(z) - zL'(z)}{L(z)^2} = \frac{1}{L(z)} \left(1 - \frac{zL'(z)}{L(z)} \right) = \frac{1 - \zeta L'(z)}{L(z)}, \quad (18)$$

where $L'(z) \equiv \frac{dL}{dz}$ represents the vertical gradient of the local Obukhov length profile. Differentiating (18) a second time with respect to z yields the vertical coordinate curvature $\zeta_{zz} \equiv \frac{\partial^2 \zeta}{\partial z^2}$:

$$\begin{aligned} \zeta_{zz} &= \frac{d}{dz} \left(\frac{L - zL'}{L^2} \right) \\ &= \frac{(L' - L' - zL'')L^2 - (L - zL') \cdot 2LL'}{L^4} \\ &= -\frac{zL''(z)}{L(z)^2} - \frac{2L'(z)}{L(z)} \left(\frac{L(z) - zL'(z)}{L(z)^2} \right) \\ &= -\frac{zL''(z)}{L(z)^2} - 2\frac{L'(z)}{L(z)}\zeta_z, \end{aligned} \quad (19)$$

which can be expressed in terms of the similarity coordinate ζ as:

$$\zeta_{zz} = -\frac{L''(z)}{L(z)}\zeta - 2\frac{L'(z)}{L(z)}\zeta_z. \quad (20)$$

Now, let $Ri_g(z) = R(\zeta(z))$ represent a general Monin–Obukhov flux-profile closure relation mapping similarity stability to the gradient Richardson number, where $R \in C^2(\mathbb{R})$. Applying the multi-variable chain rule to differentiate the physical Richardson profile $Ri_g(z)$ twice with respect to the vertical height z yields:

$$Ri_z = R_\zeta \zeta_z, \quad (21)$$

and:

$$Ri_{zz} = R_{\zeta\zeta}\zeta_z^2 + R_{\zeta}\zeta_{zz}. \quad (22)$$

Equation (22) defines the canonical GSPT curvature decomposition. It partitions the total observed vertical spatial curvature of the Richardson profile into two mathematically distinct geometric components:

$$Ri_{zz} = C_{\text{constitutive}} + C_{\text{mapping}}, \quad (23)$$

where:

$$C_{\text{constitutive}} \equiv R_{\zeta\zeta}\zeta_z^2, \quad (24)$$

$$C_{\text{mapping}} \equiv R_{\zeta}\zeta_{zz}. \quad (25)$$

The *constitutive curvature* ($C_{\text{constitutive}}$) represents the physical bending of the empirical similarity relation itself mapped into physical space. Because it scales quadratically with the coordinate Jacobian (ζ_z^2), it is highly sensitive to local coordinate stretching. The *mapping curvature* (C_{mapping}) represents the pure geometric curvature generated by the non-linear coordinate projection $z \mapsto \zeta(z)$.

2.2 Scaling Behavior at the Turning Limit ($\zeta_z \rightarrow 0$)

To understand the visual emergence of profile knees near the boundary of shear-dominated turbulent layers (such as the nose of a Low-Level Jet), we analyze the asymptotic scaling behavior of (24) and (25) as the coordinate Jacobian approaches zero ($\zeta_z \rightarrow 0$).

Assuming a standard linear flux-profile closure on the stable branch ($\phi_m(\zeta) = \phi_h(\zeta) = 1 + \beta_m\zeta$), the similarity relation maps to the gradient Richardson number via:

$$R(\zeta) = \frac{\zeta\phi_h}{\phi_m^2} = \frac{\zeta}{1 + \beta_m\zeta}. \quad (26)$$

Its respective derivatives in similarity space are given by:

$$R_{\zeta} = \frac{1}{(1 + \beta_m\zeta)^2} > 0, \quad R_{\zeta\zeta} = -\frac{2\beta_m}{(1 + \beta_m\zeta)^3}. \quad (27)$$

Substituting (27) into the GSPT partition (23) yields:

$$C_{\text{constitutive}} = -\frac{2\beta_m}{(1 + \beta_m\zeta)^3}\zeta_z^2, \quad C_{\text{mapping}} = \frac{\zeta_{zz}}{(1 + \beta_m\zeta)^2}. \quad (28)$$

Evaluating these terms in the limit of vanishing coordinate Jacobian ($\zeta_z \rightarrow 0$) under non-zero local coordinate curvature ($\zeta_{zz} = \mathcal{O}(1)$) reveals a clear scaling disparity:

$$C_{\text{constitutive}} = \mathcal{O}(\zeta_z^2), \quad (29)$$

$$C_{\text{mapping}} = \mathcal{O}(\zeta_z). \quad (30)$$

Taking the ratio of relative magnitudes under these asymptotic conditions yields:

$$\lim_{\zeta_z \rightarrow 0} \frac{|C_{\text{mapping}}|}{|C_{\text{constitutive}}|} = \lim_{\zeta_z \rightarrow 0} \frac{|R_{\zeta}\zeta_{zz}|}{|R_{\zeta\zeta}\zeta_z^2|} \propto \lim_{\zeta_z \rightarrow 0} \frac{1}{|\zeta_z|} = \infty. \quad (31)$$

Equation (31) mathematically demonstrates the coordinate dominance of relative curvature at the turning limit: as the similarity grid compresses, the observed spatial profile curvature is governed entirely by the mapping coordinate geometry rather than the underlying empirical physics of the closure function.

However, this relative divergence does *not* imply that the absolute curvature of the physical Richardson profile explodes. At a local turning point z_* where $\zeta_z(z_*) = 0$:

$$Ri_z(z_*) = R_\zeta(\zeta_*)\zeta_z(z_*) = 0, \quad (32)$$

and the second derivative simplifies to:

$$Ri_{zz}(z_*) = R_\zeta(\zeta_*)\zeta_{zz}(z_*) = \frac{\zeta_{zz}(z_*)}{(1 + \beta_m \zeta_*)^2}. \quad (33)$$

Because the similarity derivative $R_\zeta(\zeta_*)$ remains strictly positive and bounded for all finite physical stability values, the observed profile exhibits a finite, smooth, quadratic turning point ($Ri_z = 0$, $Ri_{zz} \neq \infty$) rather than a singular physical collapse.

2.3 The Linear Obukhov Profile Test

To establish a mathematical boundary between pure coordinate grid stretching and genuine vertical coordinate folding, we prove the *Linear Obukhov Profile Test*. This analytical safeguard determines whether a localized profile “knee” can be mathematically attributed to a finite-height coordinate fold.

[Linear Obukhov Profile Test] Under a strictly linear vertical profile of the Monin–Obukhov length $L(z) = L_0 + az$ (where $L_0 > 0$ represents the surface boundary value and $a = L' > 0$ represents a constant vertical flux divergence), no finite-height coordinate fold can exist, and the similarity coordinate mapping $\zeta(z)$ is globally monotone for all finite $z \geq 0$.

Assume the local Obukhov length profile is linear in z :

$$L(z) = L_0 + az, \quad L_0 > 0, \quad a > 0. \quad (34)$$

Differentiating (34) with respect to z yields the constant gradient and vanishing curvature profiles:

$$L'(z) = a, \quad L''(z) = 0. \quad (35)$$

Substituting (34) and (35) into the coordinate Jacobian equation (18) yields:

$$\zeta_z = \frac{L(z) - zL'(z)}{L(z)^2} = \frac{(L_0 + az) - az}{(L_0 + az)^2} = \frac{L_0}{(L_0 + az)^2}. \quad (36)$$

Because the surface boundary value is strictly positive ($L_0 > 0$) and the squared denominator is strictly positive for all physical heights, we have:

$$\zeta_z > 0 \quad \forall z \in [0, \infty). \quad (37)$$

Since the coordinate Jacobian is strictly positive everywhere, the mapping $z \mapsto \zeta(z)$ is globally monotone increasing. Thus, no turning point ($\zeta_z = 0$) can be realized at any finite height.

As $z \rightarrow \infty$, the similarity coordinate asymptotically compresses toward a fixed upper bound:

$$\lim_{z \rightarrow \infty} \zeta(z) = \lim_{z \rightarrow \infty} \frac{z}{L_0 + az} = \frac{1}{a}, \quad (38)$$

and the Jacobian vanishes only in the asymptotic limit:

$$\lim_{z \rightarrow \infty} \zeta_z = \lim_{z \rightarrow \infty} \frac{L_0}{(L_0 + az)^2} = 0. \quad (39)$$

Therefore, under a linear Obukhov profile, the similarity coordinate undergoes asymptotic coordinate compression rather than a finite-height fold.

The practical implication of the *Linear Obukhov Profile Test* is highly consequential for observational diagnostics: a visually sharp vertical inflection in an observed $Ri_g(z)$ profile cannot be mathematically classified as a similarity coordinate fold unless independent flux measurements establish that the local Monin–Obukhov length profile exhibits non-zero vertical curvature ($L''(z) \neq 0$).

2.4 Nondegenerate Finite-Height Folds

By utilizing the results of the *Linear Obukhov Profile Test*, we can formalize the necessary and sufficient conditions for the existence of nondegenerate, finite-height coordinate folds in physical vertical space.

[Generic Coordinate-Fold Criteria] Let $F(z) \equiv L(z) - zL'(z)$ define the coordinate turning function, such that the Jacobian is represented by $\zeta_z = \frac{F(z)}{L(z)^2}$. A nondegenerate, finite-height coordinate fold exists at a vertical coordinate $z_* > 0$ if and only if the Obukhov length profile $L(z) \in C^2$ satisfies:

$$L(z_*) = z_* L'(z_*), \quad (\text{Turning Condition})$$

$$L''(z_*) \neq 0. \quad (\text{Nondegeneracy Condition})$$

First, we establish the turning condition. For a coordinate fold to exist at z_* , the coordinate Jacobian must vanish, representing a loss of local transversality in the coordinate map:

$$\zeta_z(z_*) = 0 \iff \frac{F(z_*)}{L(z_*)^2} = 0 \iff F(z_*) = 0. \quad (40)$$

Substituting $F(z_*) = L(z_*) - z_* L'(z_*) = 0$ yields the turning condition $L(z_*) = z_* L'(z_*)$.

Second, we evaluate the nondegeneracy condition. For the coordinate fold to be nondegenerate (representing a generic quadratic turning point in the singularity-theory sense), the second spatial derivative of the mapping must remain non-zero at the turning point:

$$\zeta_{zz}(z_*) \neq 0. \quad (41)$$

Differentiating the turning function $F(z) = L(z) - zL'(z)$ with respect to z yields:

$$F'(z) = L'(z) - [L'(z) + zL''(z)] = -zL''(z). \quad (42)$$

Evaluating the second derivative of the coordinate mapping at the turning point z_* where $F(z_*) = 0$ yields:

$$\zeta_{zz}(z_*) = \frac{d}{dz} \left(\frac{F(z)}{L(z)^2} \right) \Big|_{z_*} = \frac{F'(z_*)L(z_*)^2 - F(z_*) \cdot 2L(z_*)L'(z_*)}{L(z_*)^4} = \frac{F'(z_*)}{L(z_*)^2}. \quad (43)$$

Substituting $F'(z_*) = -z_* L''(z_*)$ into (43) yields the exact coordinate curvature at the fold locus:

$$\zeta_{zz}(z_*) = -\frac{z_* L''(z_*)}{L(z_*)^2}. \quad (44)$$

For $z_* > 0$ and $L(z_*) \neq 0$, (44) proves that:

$$\zeta_{zz}(z_*) \neq 0 \iff L''(z_*) \neq 0. \quad (45)$$

This completes the proof. A nondegenerate coordinate fold requires both the local turning condition to be satisfied and the local Obukhov length profile to possess non-zero curvature ($L''(z_*) \neq 0$).

3 The Operator Non-Commutation Hazard and Preprocessing Regularization

Evaluating stability coordinates and profile curvature in the Stable Boundary Layer (SBL) relies heavily on reconstructing the vertical gradient Richardson number (Ri_g) and its vertical spatial derivatives from discrete, noisy observations [? ?]. However, because Ri_g is a non-linear

quotient of spatial gradients, the mathematical operators used for filtering, differencing, and division do not commute [? ?]. In this section, we formally analyze the spectral properties of gradient noise amplification, prove the non-commutation of the regularization and quotient operators, and demonstrate that post-diagnostic smoothing (filtering the calculated Ri_g profile) manufactures artificial coordinate folds (“knees”) near low-shear regions such as Low-Level Jet (LLJ) noses [? ?].

3.1 Spectral Gradient Noise Amplification

Differentiating discrete observational tower or profiler measurements is a classically ill-posed inverse problem [? ?]. Let $q(z) = q_{\text{true}}(z) + \epsilon(z)$ represent an observed primitive atmospheric field (e.g., u , v , or θ_v), where $q_{\text{true}}(z)$ is the smooth physical state and $\epsilon(z)$ represents high-frequency, uncorrelated instrument noise [?]. In Fourier space, the spatial derivative operator maps the spectral representation $\widehat{q}(k)$ as [?]:

$$\widehat{q}'(k) = ik\widehat{q}(k) = ik\widehat{q_{\text{true}}}(k) + ik\widehat{\epsilon}(k), \quad (46)$$

where k is the spatial wavenumber and $i = \sqrt{-1}$. Equation (46) demonstrates that taking a first-order vertical spatial derivative multiplies the noise amplitude by $|k|$ and amplifies the noise power/variance by k^2 .

For the second spatial derivative, the spectral mapping is given by [?]:

$$\widehat{q}''(k) = -k^2\widehat{q}(k) = -k^2\widehat{q_{\text{true}}}(k) - k^2\widehat{\epsilon}(k). \quad (47)$$

Here, the derivative noise amplitude scales quadratically as k^2 , while the noise power scales as k^4 .

On irregular, non-uniform boundary-layer grids (such as Cabauw or CASES-99), applying naive, unweighted local finite-difference stencils to compute vertical gradients amplifies these high-frequency spectral components [? ?]. This noise amplification frequently causes random sensor jitter to appear as spurious turbulence transitions or false gradient inversions, corrupting downstream stability analyses.

3.2 The Non-Commutation of Inversion Operators

To resolve noise amplification, we introduce three distinct mathematical operators:

1. **The Discrete Spatial Difference Operator ($D_{\Delta z}$):** A discrete operator representing vertical spatial differentiation [?].
2. **The Tikhonov–Morozov Regularization Filter (M_δ):** A smoothing operator parameterized by a regularization scale α (chosen via Morozov’s Discrepancy Principle to match the instrument noise floor δ) that maps noisy data to a smooth reconstructed profile [? ?].
3. **The Non-Linear Quotient Operator ($\mathcal{Q}[A, B] \equiv \frac{A}{B}$):** A division mapping that takes two vertical fields and evaluates their point-by-point ratio [?].

These operators exhibit two critical non-commutation properties that govern SBL profile analysis [? ?]:

[Differential-Smoothing Non-Commutation] The discrete spatial difference operator $D_{\Delta z}$ does not commute with the Tikhonov–Morozov regularization filter M_δ on arbitrary, non-uniform grids [? ?]:

$$D_{\Delta z}M_\delta \neq M_\delta D_{\Delta z}. \quad (48)$$

[Quotient-Smoothing Non-Commutation] The non-linear quotient operator $\mathcal{Q}$ does not commute with the Tikhonov–Morozov regularization filter M_δ on any non-trivial field domain [? ?]:

$$\mathcal{Q}[M_\delta A, M_\delta B] \neq M_\delta \mathcal{Q}[A, B]. \quad (49)$$

For the Richardson number calculation, this quotient non-commutation translates directly to [?]:

$$\frac{M_\delta N^2}{M_\delta S^2} \neq M_\delta \left(\frac{N^2}{S^2} \right), \quad (50)$$

where N^2 is the squared buoyancy frequency and $S^2 = u_z^2 + v_z^2$ is the squared vertical wind shear.

3.3 Spurious Fold Generation via Post-Diagnostic Smoothing

When processing campaign observations, a common methodological error is to compute a raw, noisy Richardson profile (Ri_g^{raw}) via naive finite differences and subsequently apply a smoothing filter (M_δ) to the computed output [?]. This “post-diagnostic” smoothing path is represented by the composite operator sequence:

$$Ri_g^{\text{post-smooth}} = M_\delta \circ \mathcal{Q} \circ D_{\Delta z}[u, v, \theta_v]. \quad (51)$$

The post-diagnostic path (51) manufactures spurious coordinate fold geometries due to two coupled numerical mechanisms:

3.3.1 1. Non-Linear Noise Transformation and Jump Discontinuities

The raw gradient Richardson number, $Ri_g = \frac{g}{\theta_v} \frac{\theta_{v,z}}{U_z^2}$, is a non-linear mapping of primitive variables [? ?]. This non-linear mapping transforms linear, Gaussian instrument measurement errors in u , v , and θ_v into strongly **non-Gaussian, non-linear, state-dependent noise** in Ri -space.

Under stable stratification, this state-dependent noise introduces sharp jump discontinuities in the residual discrepancy function as the smoothing parameter α varies [?]. Because of these “discrepancy jump gaps,” standard Morozov root-finding algorithms fail, as no stable α can be resolved within the classical discrepancy interval [?]. The filter is forced to toggle unstably between under-smoothing and over-smoothing, generating artificial “jumps” and knees in the filtered profile.

3.3.2 2. Low-Shear Noise Blowup

Near the nose of a Low-Level Jet, vertical wind shear weakens and approaches zero ($U_z \rightarrow 0$) [?]. Propagating primitive gradient errors ($\sigma_{U_z}^2$) into the gradient Richardson number variance yields the sensitivity relation:

$$\sigma_{Ri_g, U_z}^2 = \left(\frac{\partial Ri_g}{\partial U_z} \right)^2 \sigma_{U_z}^2 = 4 \left(\frac{g}{\theta_v} \right)^2 \frac{\theta_{v,z}^2}{U_z^6} \sigma_{U_z}^2. \quad (52)$$

Equation (52) proves that gradient uncertainty is amplified elementwise as U_z^{-6} in the local variance.

If we calculate the Richardson profile naively before smoothing, this U_z^{-6} scaling causes measurement noise to explode near zero-shear zones, creating massive unphysical spikes in the raw Ri_g profile [?]. Applying the smoothing filter M_δ afterward to these extreme spikes causes the regularized curve to “bend” artificially to satisfy the discrepancy fit, manufacturing a smooth, quadratic “knee” in physical space. This visual knee is a pure numerical artifact of the post-diagnostic filter’s struggle to smooth out low-shear singularities, rather than a physical transition or a coordinate fold of the underlying boundary layer [?].

3.4 Operational Primitive-Field Regularization

To eliminate these non-commutation artifacts and ensure that the noise model remains physically grounded at the **sensor-variable level** (where instrument errors are well-characterized), the GSPT pipeline must execute the **pre-diagnostic regularization path**:

$$Ri_g^{\text{pre-smooth}} = \mathcal{Q} \circ D_{\Delta z} \circ M_{\delta}[u, v, \theta_v]. \quad (53)$$

Under the pre-diagnostic path (53), the regularization filter M_{δ} is applied directly to the observed primitive variables prior to spatial differentiation and quotient division [? ?]. The regularized Richardson profile is constructed as:

$$Ri_g^{(\delta)}(z) = \frac{\frac{g}{M_{\delta}\theta_v(z)} D_{\Delta z}(M_{\delta}\theta_v(z))}{[D_{\Delta z}(M_{\delta}u(z))]^2 + [D_{\Delta z}(M_{\delta}v(z))]^2}, \quad (54)$$

where $M_{\delta}u$, $M_{\delta}v$, and $M_{\delta}\theta_v$ are the Tikhonov-smoothed primitive profiles, each solved independently using Morozov’s discrepancy principle based on their respective instrument standard deviations ($\sigma_U, \sigma_{\theta}$) [? ?].

By restricting the smoothing operator to the primitive fields, we ensure that:

1. The noise filtering is applied to variables where instrument errors are linear and Gaussian, avoiding non-Gaussian discrepancy jumps in Ri -space [?].
2. The analytical derivatives are computed directly from the regularized primitive splines, avoiding finite-difference discretization stencils that amplify noise near zero-shear regions [?].
3. Downstream diagnostic sensitivity (κ_{ζ}) and coordinate fold ratios remain uncoupled from visual filtering artifacts, isolating true physical SBL transitions from numerical processing distortions [?].

4 The Single-Column Model Operational Error-Partitioning Schema

A chronic issue in the development of Single-Column Models (SCMs) and numerical weather prediction (NWP) boundary-layer schemes is the practice of parameter tuning under stable conditions [?]. When a model’s simulated profiles of wind and temperature diverge from observations, developers typically adjust the empirical constants (e.g., β_m, β_h) within the Monin–Obukhov stability functions (ϕ_m, ϕ_h) to minimize the profile misfit [? ?].

Generalized Similarity Profile Theory (GSPT) demonstrates that this tuning methodology is frequently a **compensating error** [? ?]. In many cases, the physical-space curvature misfit is not driven by an incorrect local stability closure, but rather by unresolved vertical flux divergence ($L'(z) \neq 0$, $L''(z) \neq 0$) or coarse vertical grid spacing that fails to capture fine-scale tower gradients [? ?]. In this section, we formalize a closed, operationally testable spatial curvature residual partition:

$$\Delta_{\text{SCM}}(z) = \Delta_{\text{closure}}(z) + \Delta_{\text{geom}}(z) + \tau_{\Delta z}(z), \quad (55)$$

which mathematically isolates the true source of SCM profile error and prevents developers from corrupting local physical closures to compensate for unresolved flux-profile geometry or grid-spacing deficiencies [? ?].

4.1 The Total Curvature Residual (Δ_{SCM})

Let $Ri_{zz}^{\text{obs}}(z)$ represent the vertical spatial second derivative of the gradient Richardson number profile reconstructed from observations using the primitive-field Tikhonov–Morozov regularization pipeline detailed in Section III [?]. Let $Ri_{zz}^{\text{SCM}}(z)$ represent the corresponding vertical second derivative extracted from SCM output mapped onto the same vertical coordinate grid [?].

The *Total Curvature Residual* at height z is defined as:

$$\Delta_{\text{SCM}}(z) \equiv Ri_{zz}^{\text{SCM}}(z) - Ri_{zz}^{\text{obs}}(z). \quad (56)$$

To determine whether a non-zero $\Delta_{\text{SCM}}(z)$ represents a failure of the local turbulence parameterization or a geometric projection artifact, we must decompose this residual into independent, operational components [? ?].

4.2 Operational Definitions of the Partitioned Components

To make (55) mathematically testable rather than merely descriptive, each error term must be defined operationally relative to a reference coordinate mapping [?]. We define the following three geometric objects on the tower coordinate grid [?]:

$$\mathcal{G}_\zeta \equiv (\zeta(z), \zeta_z(z), \zeta_{zz}(z)), \quad (57)$$

$$\mathcal{G}_{Ri} \equiv (R_\zeta(\zeta), R_{\zeta\zeta}(\zeta), C_{\text{constitutive}}, C_{\text{mapping}}), \quad (58)$$

$$\mathcal{O}_{Ri} \equiv (Ri(z), Ri_{zz}(z), \tau_{\Delta z}(z)). \quad (59)$$

4.2.1 1. Constitutive Closure Failure (Δ_{closure})

The *Constitutive Closure Failure* measures the direct error introduced by the empirical similarity functions of the model [? ?]. It is operationally defined by projecting the model’s similarity closure relation, $R^{\text{model}}(\zeta)$, onto the **observed, well-conditioned coordinate mapping** $\mathcal{G}_\zeta^{\text{obs}}(z)$, and subtracting the observed curvature [?]:

$$\Delta_{\text{closure}}(z) \equiv \left(R_{\zeta\zeta}^{\text{model}} \left(\zeta^{\text{obs}}(z) \right) \left[\zeta_z^{\text{obs}}(z) \right]^2 + R_\zeta^{\text{model}} \left(\zeta^{\text{obs}}(z) \right) \zeta_{zz}^{\text{obs}}(z) \right) - Ri_{zz}^{\text{obs}}(z). \quad (60)$$

By keeping the coordinate geometry fixed to the observed state $(\zeta^{\text{obs}}, \zeta_z^{\text{obs}}, \zeta_{zz}^{\text{obs}})$, $\Delta_{\text{closure}}(z)$ isolates the performance of the local stability functions (e.g., comparing Businger–Dyer against Grachev et al. [?]) under identical physical forcing [?]. If $|\Delta_{\text{closure}}(z)| \approx 0$, the empirical stability functions are structurally correct.

4.2.2 2. Flux-Coordinate Geometric Distortion (Δ_{geom})

The *Flux-Coordinate Geometric Distortion* measures the error generated by the model’s failure to resolve the vertical structure of turbulent fluxes [? ?]. The vertical profiles of momentum and heat fluxes govern the local Obukhov length profile, $L(z)$, and its spatial derivatives, which in turn define the coordinate mapping curvature ζ_{zz} [? ?].

We define $\Delta_{\text{geom}}(z)$ by projecting the **observed similarity closure** $R^{\text{obs}}(\zeta)$ onto the SCM’s simulated coordinate mapping $\mathcal{G}_\zeta^{\text{model}}(z)$, and subtracting the observed reference mapping [?]:

$$\Delta_{\text{geom}}(z) \equiv \left(R_{\zeta\zeta}^{\text{obs}} \left(\zeta^{\text{model}}(z) \right) \left[\zeta_z^{\text{model}}(z) \right]^2 + R_\zeta^{\text{obs}} \left(\zeta^{\text{model}}(z) \right) \zeta_{zz}^{\text{model}}(z) \right) - \left(C_{\text{constitutive}}^{\text{obs}}(z) + C_{\text{mapping}}^{\text{obs}}(z) \right), \quad (61)$$

which, near coordinate turning points where mapping curvature dominates ($C_{\text{mapping}} \gg C_{\text{constitutive}}$), simplifies to the GSPT transition-surface difference [? ?]:

$$\Delta_{\text{geom}}(z) \approx \mathcal{R}_{\text{coord}}^{\text{model}}(z) - \mathcal{R}_{\text{coord}}^{\text{obs}}(z), \quad (62)$$

where $\mathcal{R}_{\text{coord}} \equiv R_\zeta \zeta_{zz}$ represents the mapping curvature profile [? ?]. If $|\Delta_{\text{geom}}(z)| \gg 0$, it indicates that the model’s simulated vertical flux divergence or mixing length profile is over-smoothed, distorting the coordinate grid transformation and shifting the height of the coordinate fold locus (z_{fold}) [?].

4.2.3 3. Discretization and Grid-Scale Artifact ($\tau_{\Delta z}$)

Coarse vertical grid spacing in SCMs introduces systematic truncation errors when approximating continuous spatial derivatives [? ?]. The *Discretization Artifact* is the residual difference between the discrete approximation of curvature on the model’s vertical grid and its continuum limit resolved on the high-resolution tower grid [? ?]:

$$\tau_{\Delta z}(z) \equiv Ri_{zz}^{\Delta z}(z) - Ri_{zz}^{\text{continuum}}(z), \quad (63)$$

where $Ri_{zz}^{\Delta z}(z)$ is evaluated using the model’s discrete spatial difference operator $D_{2,\Delta z}$ [?]. We formalize the localized truncation error using a leading-order Taylor expansion on a vertical grid spacing of Δz [?]:

$$\tau_{\Delta z}(z) = \frac{(\Delta z)^2}{12} \frac{\partial^4 Ri_g}{\partial z^4} + \mathcal{O}((\Delta z)^4). \quad (64)$$

By computing $\tau_{\Delta z}(z)$ directly from the fourth spatial derivative of our regularized primitive splines, we can quantitatively assess whether a curvature misfit is simply a numerical grid-resolution artifact rather than a physical parameterization error [?].

4.3 Operational Decision Matrix and Corrective SCM Actions

By evaluating the relative magnitudes of the partitioned residuals, SCM developers can systematically diagnose model deficiencies and execute targeted, physically consistent corrective actions without invoking compensating parameter tuning [? ?]:

5 Empirical Validation and Observational Benchmarks

To validate the pre-diagnostic regularization pipeline (53) and operationalize the SCM curvature error partition (55), we evaluate high-resolution nocturnal boundary layer profiles from the CASES-99 [?] and SHEBA [? ?] field campaigns, alongside Single-Column Model output from the GABLS3 intercomparison [?].

5.1 Elimination of Synthetic Knees in CASES-99 Tower Profiles

Applying post-diagnostic smoothing (51) to raw CASES-99 tower measurements during strong stability events ($z/L > 1$) consistently manufactures artificial coordinate fold inflections in physical vertical space near Low-Level Jet (LLJ) noses [? ?].

During October 20, 1999 (04:00 UTC), a pronounced nocturnal LLJ core formed at $z \approx 45$ m with vertical velocity shear vanishing ($U_z \rightarrow 0$) [?]. Naive spatial differencing on this sparse 6-level tower grid introduces massive noise amplification governed by the non-linear U_z^{-6} absolute variance scaling in (52), driving the local gradient Richardson number variance to $\sigma_{Ri_g}^2 > 10^3 \text{ m}^{-2}$ near the jet nose [? ?].

[Figure 5.1: Comparative Regularization Trajectories on CASES-99 Tower Profile]

Figure 1: Comparison of post-diagnostic smoothing (manufacturing an artificial quadratic knee at $z = 42$ m) versus pre-diagnostic primitive field regularization (preserving monotonic stability structure) for the CASES-99 55-m main tower.

Table 1: GSPT Operational Decision Matrix for SCM Development.

Residual Signature	Physical Interpretation	Corrective SCM Action
$ \Delta_{\text{closure}} \gg 0$ $\Delta_{\text{geom}} \approx 0$	Constitutive Closure Failure: The empirical local stability functions are physically inaccurate under stable stratification [?].	Swap the SBL boundary-layer closure scheme. Transition from standard Businger–Dyer functions to non-singular Grachev et al. [?] physics.
$ \Delta_{\text{geom}} \gg 0$ $\Delta_{\text{closure}} \approx 0$	Flux-Coordinate Distortion: The local stability functions are correct, but the model over-smooths heat and momentum flux divergence profiles (L', L'') [?].	Do not touch ϕ_m or ϕ_h. Retune the vertical mixing lengths (l_m), sub-grid TKE dissipation rates, or non-local transport closures to restore correct flux-divergence curvature [?].
$ \tau_{\Delta z} \gg 0$	Discretization Grid Artifact: Coarse model grid spacing fails to capture fine-scale vertical gradient non-linearities near the ground [?].	Refine the vertical grid layout. Apply logarithmic grid stretching in the near-surface layer ($z < 20$ m) to resolve intense gradients without altering physical equations [?].
Both Δ_{closure} and $\Delta_{\text{geom}} \gg 0$	Coupled Closure & Coordinate Error: The model suffers from both inaccurate local stability scaling and highly distorted vertical transport profiles [?].	Execute a staged calibration. First, resolve the vertical grid and flux-divergence profiles ($\Delta_{\text{geom}} \rightarrow 0$); second, calibrate the remaining constitutive stability functions [?].

When Morozov Tikhonov filtering M_δ is executed on the primitive wind and temperature fields $[u, v, \theta_v]$ prior to quotient division (54), the primitive noise levels ($\sigma_U \approx 0.05 \text{ m s}^{-1}$, $\sigma_\theta \approx 0.02 \text{ K}$) remain linear and well-conditioned [? ?]. The resulting regularized profile $Ri_g^{(\delta)}(z)$ successfully eliminates the synthetic knee, confirming that the apparent physical-space inflection is an observation-space coordinate-mapping feature (Fold Illusion) generated by low-shear coordinate compression, rather than a genuine loss of normal hyperbolicity on the fast-slow TKE equilibrium manifold [? ?].

5.2 GABLS3 Error Decomposition and Compensating Tuning

We apply the operational decision matrix (Table 1) to a 12-level SCM simulation of the GABLS3 nocturnal case study to demonstrate how GSPT isolates physical from numerical error modes [?]. Traditional evaluation yields a total curvature misfit of $\Delta_{\text{SCM}}(z) = 4.20 \text{ m}^{-2}$ near the nocturnal inversion layer ($z = 80 \text{ m}$) [?].

Table 2: Operational Error Partitioning for GABLS3 SCM Benchmark ($z = 80 \text{ m}$)

Component	Absolute Value (m^{-2})	Fractional Contribution (%)
Total Residual (Δ_{SCM})	4.20	100.0%
Constitutive Failure (Δ_{closure})	0.31	7.4%
Flux-Coordinate Distortion (Δ_{geom})	3.12	74.3%
Discretization Artifact ($\tau_{\Delta z}$)	0.77	18.3%

The partition reveals that 74.3% of the profile misfit stems from flux-coordinate geometric distortion (Δ_{geom}), driven by over-smoothed sub-grid TKE dissipation rates that suppress vertical flux-divergence curvature ($L''(z) \neq 0$) [? ?]. The constitutive stability closure failure (Δ_{closure}) contributes less than 8% to the overall misfit. Under the GSPT operational decision matrix (Table 1), this dominant geometric signature instructs model developers to retune vertical mixing length profiles (l_m) or sub-grid transport closures rather than corrupting the local empirical similarity functions (ϕ_m, ϕ_h), successfully preventing unphysical compensating calibrations [? ?].

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Author Contributions

D. England: Conceptualization, Formal Analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **[Co-Author Name]:** Investigation, Validation, Data Curation, Writing – review & editing.

Data Availability Statement

The observational datasets and open-source software used in this study are available from the following public repositories:

- **SHEBA Met City ASFG Data:** High-resolution 5-level tower measurements are archived at the NSF NCAR Earth Observing Laboratory (<https://doi.org/10.5065/D65H7DNS>) [?].
- **CASES-99 Dataset:** Nocturnal boundary-layer tower and surface flux observations are accessible via the NCAR EOL Data Archive [?].
- **GABLS3 Benchmark Data:** Single-column model forcing profiles and tower validation sets are maintained by the Royal Netherlands Meteorological Institute (KNMI) [?].
- **Software Infrastructure:** The Julia numerical pipeline for Generalized Similarity Profile Theory (GSPT), Tikhonov–Morozov regularization, and spatial curvature partitioning is open-source and hosted on GitHub (<https://github.com/sbltoolkit/SBLToolkit.jl>) [?].

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